Analytic Interdomain Kernel Construction for One-Dimensional Online HiPPO Sparse Variational Gaussian Processes

Abstract

Online Gaussian processes (GPs) are attractive for streaming prediction because they combine flexible function modeling with calibrated uncertainty, but exact inference is cubic and standard sparse online updates can forget early observations. Online HiPPO Sparse Variational Gaussian Processes (OHSVGP) addresses this issue by using HiPPO-based interdomain inducing variables, yet its interdomain kernel construction is built from temporal ODE recursion of Fourier–Legendre states. We study an analytic replacement for this kernel-construction step in the one-dimensional HiPPO-LegS setting with stationary kernels. The main idea is to express the required sine/cosine–Legendre overlap integrals in closed form through spherical Bessel functions, so that kernel quantities are evaluated directly at time t and the remaining recurrence is over polynomial order rather than over timestamps. This changes the recursion structure of interdomain kernel construction, removes ODE discretization error from that construction, and reduces its asymptotic dependence on stream length. On Solar Irradiance and COVID-19 mortality streams, the resulting analytic variant maintains comparable predictive performance and is slightly favorable on several aggregate metrics relative to ODE-based OHSVGP. The present derivation is specific to one-dimensional HiPPO-LegS with stationary kernels, and dedicated runtime scaling experiments remain future work.

1. Introduction

Gaussian processes (GPs) remain a standard probabilistic tool for time-series modeling because they combine nonparametric flexibility with principled uncertainty quantification (Roberts et al., 2013). Their deployment in online settings, however, is limited by the cubic complexity of exact inference and by the fact that sparse online approximations (Titsias, 2009; Hensman et al., 2013; Bui et al., 2017; Maddox et al., 2021) often struggle with catastrophic forgetting.

The Online HiPPO Sparse Variational Gaussian Process (OHSVGP) framework (Chen et al., 2025) addresses this issue by constructing the inducing variables for GPs with high-order polynomial projection operators (HiPPO) (Gu et al., 2020), substantially alleviating catastrophic forgetting in online regression and continual-learning settings. By defining inducing features as HiPPO's time-varying orthogonal polynomials over $[0, t]$, a fixed set of inducing variables optimally compresses the entire observation history as the horizon grows.

Despite these benefits, the original implementation relies on ODE-based evolution of the interdomain features, which requires discretization to integrate over time. Therefore, as time series become longer, integration becomes computationally expensive and discretization errors accumulate. We observe that for HiPPO-LegS in one dimension, the sine/cosine–Legendre inner products required by the interdomain construction admit closed forms via spherical Bessel functions, removing the need for ODE recursion altogether.

Our contributions are threefold. First, we show that the sine/cosine–Legendre integrals underlying the OHSVGP interdomain kernels admit closed forms via spherical Bessel functions, yielding analytic expressions for $K_{fu}(t)$ and $K_{uu}(t)$ that replace the ODE solver entirely. Second, we demonstrate that the analytic construction avoids per-step ODE recursion, reducing kernel cost significantly when the stream length far exceeds the basis size, and eliminates the discretization error that accumulates over long horizons. Third, we validate on Solar Irradiance and COVID-19 mortality streams that the analytic variant matches or improves ODE-based OHSVGP in predictive accuracy.

2. Background

Here we review the necessary background on GPs and their sparse approximations, and interdomain GPs, then introduce the online HiPPO sparse GP framework which motivates our analytic extension.

2.1. Gaussian Processes and Sparse Variational Inference

Let $f \sim \mathcal{GP}(0, k)$ be a Gaussian process with covariance kernel k. Given data $\{(x_i, y_i)\}_{i=1}^{N}$, exact GP inference

scales cubically in N, motivating sparse variational approximations based on inducing variables u (Titsias, 2009; Hensman et al., 2013). In the standard formulation, the variational posterior is defined as

$$q(f, u) = p(f \mid u)q(u), \tag{1}$$

where $q(u)$ is usually Gaussian with free mean and covariance. For Gaussian likelihoods, collapsed objectives are available; for non-conjugate likelihoods, stochastic variational inference is typically used.

Interdomain GPs generalize inducing variables from point evaluations to integral transforms,

$$u_m = \int f(x)\phi_m(x)\,dx, \tag{2}$$

where $\{\phi_m\}$ are basis functions chosen to induce desirable structure (Lázaro-Gredilla and Figueiras-Vidal, 2009; van der Wilk et al., 2020). The corresponding covariances are

$$[K_{fu}]_m = \int k(x, x')\phi_m(x')\,dx', \tag{3}$$

$$[K_{uu}]_{nm} = \iint k(x, x')\phi_n(x)\phi_m(x')\,dx\,dx'. \tag{4}$$

2.2. Online Sparse Gaussian Processes

In online learning, data arrive sequentially. Standard online sparse variational GP methods update the posterior using only new data and the previous variational approximation (Bui et al., 2017). While computationally attractive, this approach does not guarantee memory retention, as inducing locations remain optimized relative to recent observations.

2.3. HiPPO and Recurrent Memory

HiPPO constructs a finite-dimensional memory state by projecting historical signals onto time-varying orthogonal polynomial bases (Gu et al., 2020). For HiPPO-LegS, the basis functions are scaled Legendre polynomials over the interval $[0, t]$, producing coefficients that satisfy a linear ODE:

$$\frac{d}{dt}c(t) = A(t)c(t) + B(t)f(t), \tag{5}$$

supporting streaming updates and long-range memory.

2.4. Online HiPPO Sparse Variational Gaussian Processes

OHSVGP turns HiPPO coefficients into interdomain inducing variables,

$$u_m(t) = \int f(x)\phi_m^{(t)}(x)\,dx, \tag{6}$$

with time-dependent basis $\phi_m^{(t)}$. In the one-dimensional HiPPO-LegS case considered here, the basis functions on $[0, t]$ are

$$\phi_\ell^{(t)}(x) = \frac{2\ell+1}{t} P_\ell\left(\frac{2x}{t} - 1\right)\mathbf{1}_{[0,t]}(x), \tag{7}$$

which induce the interdomain quantities

$$[K_{fu}(t)]_{n\ell} = \int_0^t k(x_n, x)\phi_\ell^{(t)}(x)\,dx, \tag{8}$$

$$[K_{uu}(t)]_{\ell m} = \int_0^t \int_0^t k(x, x')\phi_\ell^{(t)}(x)\phi_m^{(t)}(x')\,dx\,dx'. \tag{9}$$

In the original OHSVGP implementation, these kernel terms are built in Random Fourier Feature (RFF) space (Rahimi and Recht, 2007) by evolving Fourier-Legendre coefficients with a HiPPO ODE over time. This makes the feature matrix $Z(t)$ available online, but also ties kernel construction to a time-stepping procedure whose cost and discretization error both grow with stream length.

3. Methods

We now show that, in the one-dimensional HiPPO-LegS setting with a stationary kernel, the interdomain kernels $K_{fu}(t)$ and $K_{uu}(t)$ can be evaluated at a target time t in closed form, without the temporal HiPPO ODE recursion that the baseline OHSVGP uses to construct them. The derivation uses two well-known tools. Bochner's theorem factors a stationary kernel into a spectral average over Fourier features, and the plane-wave/Legendre identity expresses integrals of Legendre polynomials against a plane wave in terms of spherical Bessel functions. Combining them, every entry of the RFF feature matrix $Z(t)$ underlying $K_{fu}(t)$ and $K_{uu}(t)$ becomes a scalar involving $j_\ell(\omega t/2)$, and the sequence $\{j_\ell\}$ is generated by a short recursion over polynomial order rather than over time.

We develop the construction in three steps. Section 3.1 reduces the kernel integrals to one-dimensional sine/cosine projections of the HiPPO-LegS basis via Bochner's theorem. Section 3.2 derives the closed-form expression of each projection via the plane-wave/Legendre identity. Section 3.3 describes the polynomial-order recursion used to evaluate the resulting spherical Bessel sequence and summarizes the computational cost. Figure 1 contrasts the full construction with the baseline ODE-based assembly.

3.1. Bochner Factorization and the Overlap Integrals

We first restate the setup we will use. For one-dimensional input $x = t$ and basis size M, the HiPPO-LegS basis on $[0, t]$ is defined, for $\ell = 0, \ldots, M-1$, by

$$\phi_\ell^{(t)}(x) = \frac{2\ell+1}{t} P_\ell\left(\frac{2x}{t} - 1\right)\mathbf{1}_{[0,t]}(x), \tag{10}$$

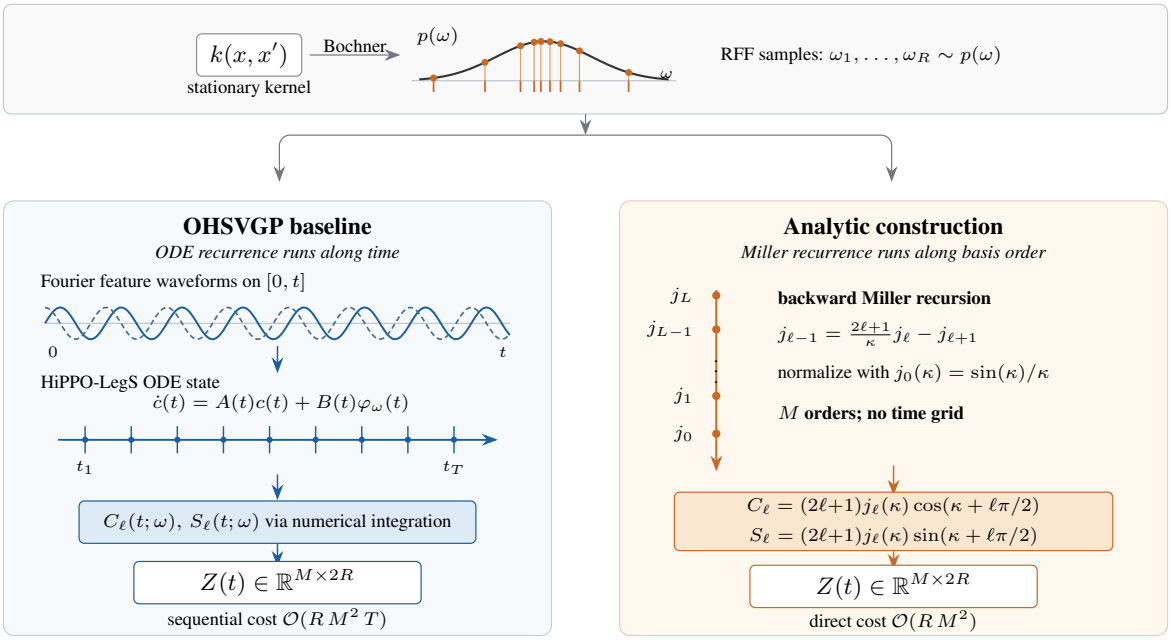


Figure 1. Two ways to assemble the HiPPO-LegS interdomain features $Z(t)$ in RFF space. Both start from the Bochner sampling of R frequencies $\{\omega_r\} \sim p(\omega)$. OHSVGP (left, blue) evaluates the sine/cosine–Legendre inner products by feeding the Fourier feature waveforms through an M-dimensional HiPPO-LegS ODE whose state is advanced at every discretization step, so the recursion runs along time and costs $\mathcal{O}(RM^2T)$. Our analytic construction (right, orange) instead runs a backward Miller recursion along the basis order ℓ (starting from a seed at $\ell = L \gg M$ and descending to $\ell = 0$), which returns all spherical Bessel values in $\mathcal{O}(M)$ per frequency; the projections C_ℓ, S_ℓ then follow in closed form. The axis of recursion therefore switches from time t to basis order ℓ, eliminating both the T factor and the accumulated discretization error.

where P_ℓ is the Legendre polynomial of degree ℓ and $\mathbf{1}_{[0,t]}$ is the indicator of $[0, t]$. Given N observation locations $\{x_n\}_{n=1}^N$, the interdomain cross-covariance $K_{fu}(t) \in \mathbb{R}^{N\times M}$ and the inducing covariance $K_{uu}(t) \in \mathbb{R}^{M\times M}$ have entries

$$[K_{fu}(t)]_{n\ell} = \int_0^t k(x_n, x)\, \phi_\ell^{(t)}(x)\, dx, \tag{11}$$

$$[K_{uu}(t)]_{\ell m} = \int_0^t \int_0^t k(x, x')\, \phi_\ell^{(t)}(x) \phi_m^{(t)}(x')\, dx\, dx'. \tag{12}$$

We assume throughout that the kernel k is stationary and admits a spectral density $p(\omega)$, so that Bochner's theorem gives

$$k(x, x') = \mathbb{E}_{\omega\sim p(\omega)}\big[\cos(\omega x)\cos(\omega x') + \sin(\omega x)\sin(\omega x')\big]. \tag{13}$$

Drawing R independent frequencies $\{\omega_r\}_{r=1}^R \sim p(\omega)$ replaces the expectation by an empirical average. Substituting this factorization into $K_{fu}(t)$ and $K_{uu}(t)$ and exchanging expectation with integration over x shows that each sampled frequency ω contributes to the kernels only through a pair of one-dimensional projections of the HiPPO-LegS basis:

$$C_\ell(t;\omega) = \int_0^t \cos(\omega x)\, \phi_\ell^{(t)}(x)\, dx, \tag{14}$$

$$S_\ell(t;\omega) = \int_0^t \sin(\omega x)\, \phi_\ell^{(t)}(x)\, dx. \tag{15}$$

The computational problem therefore reduces to evaluating the $2M$ scalars $\{C_\ell(t;\omega_r), S_\ell(t;\omega_r)\}_{\ell=0}^{M-1}$ at a target time t, for each $r = 1, \ldots, R$. These are the overlaps we evaluate in closed form next.

3.2. Closed-Form via Spherical Bessel Functions

The key observation is that a change of variables to $u = 2x/t - 1$ turns C_ℓ and S_ℓ into the classical plane-wave/Legendre integral, which is known in closed form via spherical Bessel functions. The choice of u matches the argument of the Legendre polynomial in $\phi_\ell^{(t)}$, which already has the form $P_\ell(2x/t - 1)$, and maps $[0, t]$ to the standard Legendre domain $[-1, 1]$.

We combine C_ℓ and S_ℓ into a single complex integral

$$C_\ell(t;\omega) + i\, S_\ell(t;\omega) = \int_0^t e^{i\omega x}\, \phi_\ell^{(t)}(x)\, dx, \tag{16}$$

and introduce

$$u = \frac{2x}{t} - 1, \qquad \kappa = \frac{\omega t}{2}. \tag{17}$$

After substitution, the Legendre polynomial depends on u, and the exponential becomes a plane wave $e^{i\omega x} =$

$e^{i\kappa(u+1)} = e^{i\kappa}\, e^{i\kappa u}$. Applying the classical plane-wave/Legendre identity (DLMF)

$$\int_{-1}^{1} e^{i\kappa u}\, P_\ell(u)\, du = 2\, i^\ell\, j_\ell(\kappa), \tag{18}$$

where j_ℓ is the spherical Bessel function of the first kind, closes the remaining integral. Factoring out $e^{i\kappa}$ and using $i^\ell = \cos(\ell\pi/2) + i\sin(\ell\pi/2)$, the real and imaginary parts give the closed-form overlaps

$$C_\ell(t;\omega) = (2\ell+1)\, j_\ell(\kappa)\, \cos(\kappa + \ell\pi/2), \tag{19}$$

$$S_\ell(t;\omega) = (2\ell+1)\, j_\ell(\kappa)\, \sin(\kappa + \ell\pi/2), \tag{20}$$

with $\kappa = \omega t/2$.

In these closed forms, t enters only through the scalar $\kappa = \omega t/2$; the polynomial order ℓ enters through the Bessel index and the phase shift. Stacking the $2M$ scalars $\{C_\ell(t;\omega_r), S_\ell(t;\omega_r)\}_{\ell=0}^{M-1}$ over the R sampled frequencies yields the RFF feature matrix $Z(t) \in \mathbb{R}^{M\times 2R}$ of the baseline OHSVGP. Writing $Z_f(X) \in \mathbb{R}^{N\times 2R}$ for the corresponding data-side random feature matrix, the interdomain kernels follow as

$$K_{uu}(t) \approx Z(t)\, Z(t)^\top, \tag{21}$$

$$K_{fu}(t) \approx Z_f(X)\, Z(t)^\top. \tag{22}$$

We obtain the same feature matrix $Z(t)$ as the ODE baseline, but computed directly at t instead of through time stepping.

3.3. Numerical Evaluation and Complexity

The only remaining computational task is the evaluation of $\{j_\ell(\kappa)\}_{\ell=0}^{M-1}$ for each sampled frequency. The three-term recurrence

$$j_{\ell-1}(\kappa) = \frac{2\ell+1}{\kappa}\, j_\ell(\kappa) - j_{\ell+1}(\kappa) \tag{23}$$

is numerically unstable when applied upward from j_0, j_1 once ℓ exceeds κ, because it amplifies the irregular Bessel solution. We therefore evaluate it by backward Miller recursion (Gautschi, 1967): starting from a sufficiently large order $L \gg M$ with an arbitrary non-zero seed pair (j_L, j_{L+1}), we apply the recurrence downward to $\ell = 0$ and rescale the whole sequence so that $j_0(\kappa) = \sin(\kappa)/\kappa$ matches its known value. For small κ, Taylor expansions around the origin avoid division-by-zero in this normalization. The procedure is purely arithmetic and compatible with auto-differentiation, and it yields all M orders in $O(M)$ work per frequency. [Miller recursion has its own truncation error; the "removes discretization error" claim needs empirical backing (e.g., a Δt ablation showing that the ODE baseline approaches the analytic result as $\Delta t \to 0$).]

For a stream of length T with basis size M and R sampled frequencies, the ODE-based baseline performs an $O(RM^2)$ update at each of the T discretization steps, giving a total cost of $O(RM^2T)$. The analytic construction evaluates $K_{fu}(t)$ and $K_{uu}(t)$ directly at any target t in $O(RM^2)$, independent of the number of preceding steps. [Cost reduction is only a Big-O argument; no runtime timing is reported. §4.4 acknowledges the missing scaling study.]

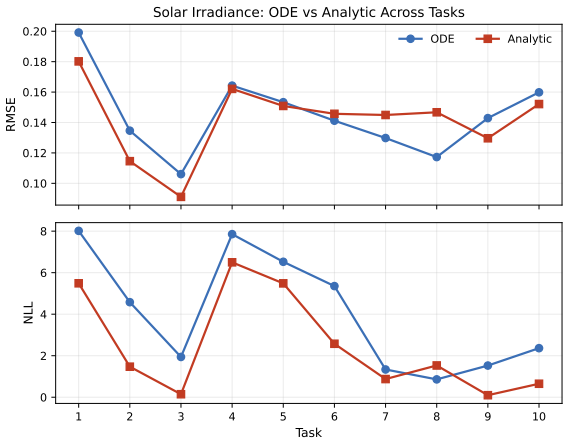


Figure 2. Task-wise RMSE and NLL across the 10-task solar stream.

4. Experiments

We evaluate the analytic construction against the ODE baseline on two streaming benchmarks: solar irradiance with a Gaussian likelihood and COVID-19 mortality with a Negative Binomial likelihood.

4.1. Solar Irradiance

We evaluate a matched analytic-versus-ODE ablation of the OHSVGP framework on 10 sequential solar irradiance tasks with a Gaussian likelihood. The two variants share the same data split, kernel family, initialization, inducing budget, and spectral budget. We report task-wise RMSE and predictive negative log-likelihood (NLL), and defer representative posterior plots to Appendix A.

Table 1 summarizes the solar ablation over the 10-task stream. On this benchmark, the analytic construction reduces mean RMSE from 0.145 to 0.142 and mean NLL from 4.035 to 2.482. The analytic model achieves lower RMSE on 7/10 tasks and lower NLL on 9/10 tasks, with the largest gain on Task 2 and a clear regression on Task 8.

Table 1. Solar irradiance comparison across 10 sequential tasks. Lower is better. Full task-wise values are given in Appendix A.

Method	RMSE ↓	NLL ↓	RMSE wins ↑	NLL wins ↑
ODE	0.145 ± 0.025	4.035 ± 2.637	3/10	1/10
Analytic	$\mathbf{0.142 \pm 0.024}$	$\mathbf{2.482 \pm 2.307}$	**7/10**	**9/10**

Figure 2 visualizes the stream-level metric trends, while Appendix A gives representative posterior comparisons. Tasks 1 and 10 illustrate that the analytic variant remains competitive in both early and late stages of the stream, while Task 2 highlights the largest improvement in both RMSE and NLL. We also include Task 8 as a failure case, where the ODE variant outperforms the analytic approximation.

4.2. COVID-19 Mortality

We next evaluate the same matched analytic-versus-ODE comparison on 5 sequential COVID-19 mortality tasks with a Negative Binomial likelihood. Here we summarize all 15 seen-task evaluation pairs induced by the 5-task stream. Our primary metrics are negative log predictive density (NLPD) and expected calibration error (ECE), while RMSE and R^2 are reported as secondary diagnostics in Appendix B.

Table 2. COVID-19 mortality comparison across all 15 seen-task evaluation pairs. Lower is better. Full task-wise values are given in Appendix B.

Method	NLPD ↓	ECE ↓	NLPD wins ↑	ECE wins ↑
ODE	3.548 ± 0.410	0.109 ± 0.041	4/15	8/15
Analytic	$\mathbf{3.539 \pm 0.430}$	$\mathbf{0.109 \pm 0.038}$	**11/15**	7/15

Table 2 reports the COVID comparison. Averaged over all 15 seen-task evaluation pairs, the analytic variant yields slightly lower mean NLPD (3.539 vs. 3.548) and essentially matching mean ECE (both ≈ 0.109). It achieves lower NLPD on 11/15 pairs and lower ECE on 7/15 pairs, indicating modest aggregate gains in probabilistic accuracy. The final stage after Task 5 is more mixed: the analytic model still attains slightly lower mean NLPD (3.714 vs. 3.719) and slightly higher mean R^2 (0.701 vs. 0.699), but its mean ECE worsens from 0.101 to 0.114 because of a clear regression on the final held-out Task 5 evaluation. Overall, these results suggest that the analytic construction maintains predictive behavior comparable to ODE-based OHSVGP on this continual setting.

Appendix B visualizes the task-pair trajectories for RMSE, R^2, NLPD, and ECE across the full stream. The analytic variant tracks the ODE baseline closely in RMSE and R^2, with its clearest probabilistic gain appearing on Eval 1 after Task 4, where both NLPD and ECE improve noticeably. The main counterexample occurs on the final Eval 5 after Task 5 pair, which explains the weaker final-stage ECE despite the stronger aggregate NLPD trend.

[Reads as if the appendix figure is assumed to be in front of the reader.]

4.3. Efficiency and Scaling Evaluation

TODO (runtime and scaling study). The complexity discussion in §3 predicts that analytic kernel construction should reduce the asymptotic dependence on stream length from $O(RM^2T)$ to $O(RM^2)$ for fixed evaluation time. A dedicated runtime study is still needed to test whether this theoretical change translates into practical gains on the current implementation.

TODO (evaluation protocol). The intended empirical checks are: kernel-construction time versus stream length T; kernel-construction time versus basis size M; optionally runtime versus number of spectral features R; total online update time; and, if implementation details permit, memory usage. We do not report such measurements here because they are not yet available in the current source.

5. Conclusion and Future Work

We presented an analytic replacement for ODE-based interdomain kernel construction in one-dimensional HiPPO-LegS OHSVGP. The central change is a shift from temporal ODE recursion to direct kernel evaluation together with recursion over polynomial order, implemented through spherical Bessel identities. Within this restricted setting, the resulting construction removes ODE discretization error from interdomain kernel construction and reduces its asymptotic dependence on stream length, while maintaining or slightly improving predictive performance on the evaluated benchmarks. Important next steps are to test the efficiency implications with dedicated runtime scaling experiments, to study alternative basis functions and spectral approximations, and to determine whether analogous analytic constructions are possible in higher-dimensional settings.

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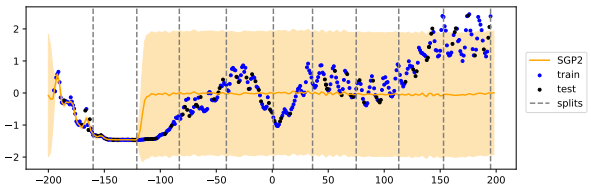
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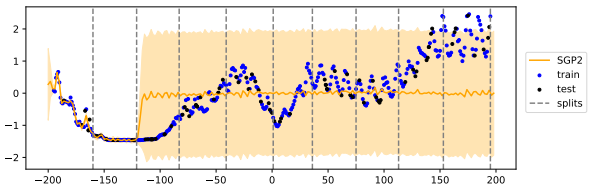
A. Additional Solar Results

Table 3. Task-wise solar irradiance comparison between the ODE-based and analytic variants. Negative deltas indicate an improvement of the analytic method over the ODE baseline.

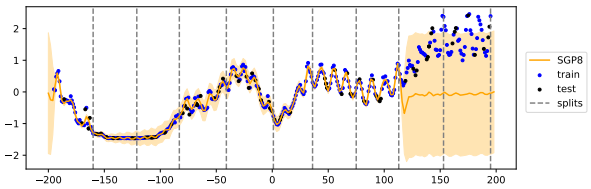
Task	ODE RMSE	Analytic RMSE	Δ RMSE	ODE NLL	Analytic NLL	Δ NLL
1	0.1992	0.1802	-0.0190	8.0143	5.4882	-2.5261
2	0.1346	0.1146	-0.0201	4.5787	1.4713	-3.1074
3	0.1060	0.0911	-0.0149	1.9400	0.1451	-1.7949
4	0.1643	0.1621	-0.0021	7.8561	6.4982	-1.3578
5	0.1534	0.1509	-0.0025	6.5244	5.4849	-1.0395
6	0.1412	0.1457	0.0045	5.3541	2.5755	-2.7786
7	0.1298	0.1449	0.0152	1.3365	0.8754	-0.4611
8	0.1173	0.1467	0.0294	0.8595	1.5299	0.6704
9	0.1429	0.1296	-0.0133	1.5256	0.0976	-1.4281
10	0.1598	0.1521	-0.0077	2.3635	0.6493	-1.7143
Mean	0.1448	0.1418	-0.0030	4.0353	2.4815	-1.5537



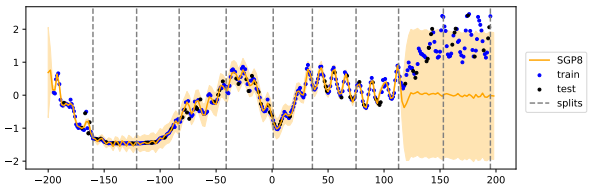
(a) Task 2, ODE



(b) Task 2, Analytic

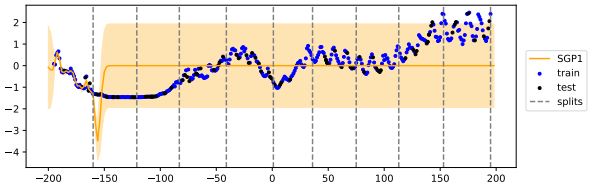


(c) Task 8, ODE

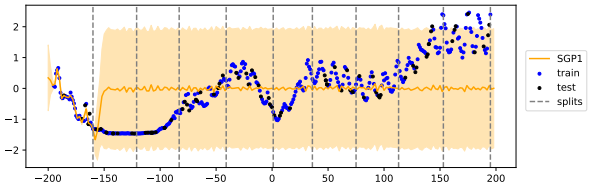


(d) Task 8, Analytic

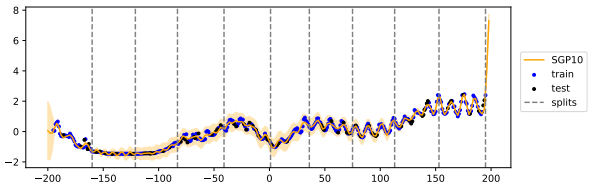
Figure 3. Representative solar posterior comparisons. Task 2 shows the strongest improvement of the analytic construction in both RMSE and NLL, while Task 8 provides the main counterexample where the ODE construction performs better.



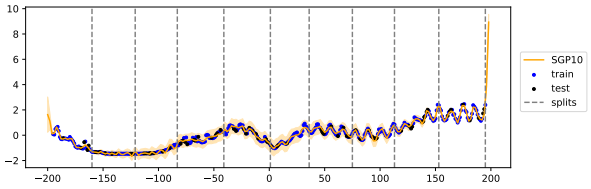
(a) Task 1, ODE



(b) Task 1, Analytic



(c) Task 10, ODE



(d) Task 10, Analytic

Figure 4. Early- and late-stream posterior comparisons on solar irradiance. The analytic construction remains competitive at the beginning of the stream and retains an advantage in the later stage, consistent with the improved aggregate NLL.

B. Additional COVID Results

Table 4. Task-wise COVID-19 mortality comparison between the ODE-based and analytic variants. Negative deltas indicate an improvement of the analytic method over the ODE baseline.

Eval	After	ODE NLPD	Analytic NLPD	Δ NLPD	ODE ECE	Analytic ECE	Δ ECE
1	1	2.9599	2.9636	0.0037	0.1500	0.1463	-0.0037
1	2	2.9783	2.9489	-0.0294	0.1278	0.1278	0.0000
2	2	3.7922	3.7910	-0.0011	0.0800	0.0717	-0.0083
1	3	2.9806	2.9527	-0.0279	0.1352	0.1352	0.0000
2	3	3.6662	3.6625	-0.0037	0.0767	0.0767	0.0000
3	3	3.8487	3.8628	0.0141	0.1183	0.1117	-0.0067
1	4	3.0697	2.9829	-0.0869	0.1907	0.1426	-0.0481
2	4	3.6496	3.6491	-0.0006	0.0767	0.0867	0.0100
3	4	3.8816	3.8965	0.0149	0.1283	0.1200	-0.0083
4	4	3.8058	3.8005	-0.0053	0.0517	0.0433	-0.0083
1	5	3.0870	3.0674	-0.0195	0.1537	0.1426	-0.0111
2	5	3.6310	3.6207	-0.0102	0.0867	0.1183	0.0317
3	5	3.8031	3.7754	-0.0277	0.0750	0.0767	0.0017
4	5	3.9312	3.9027	-0.0285	0.0533	0.0583	0.0050
5	5	4.1404	4.2054	0.0650	0.1375	0.1750	0.0375
Overall mean		3.5484	3.5388	-0.0095	0.1094	0.1089	-0.0006
Final mean		3.7185	3.7143	-0.0042	0.1012	0.1142	0.0129

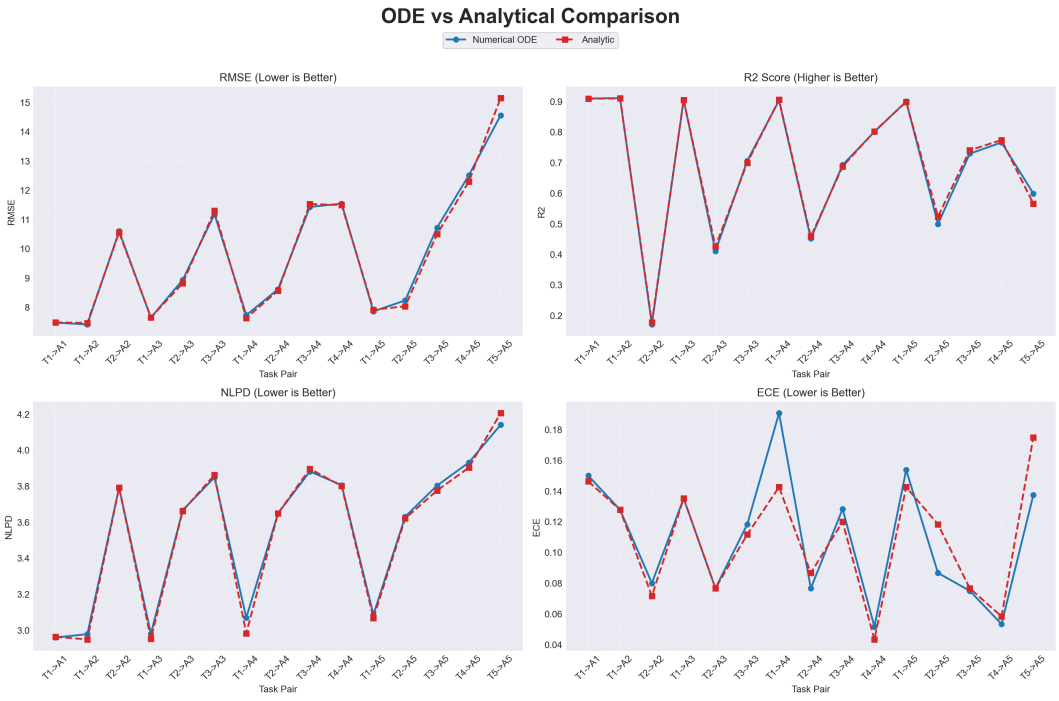
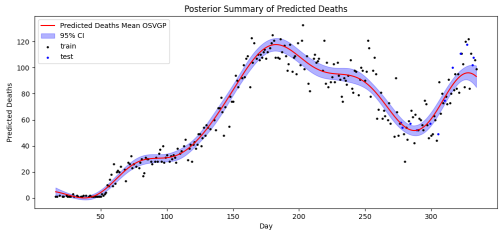
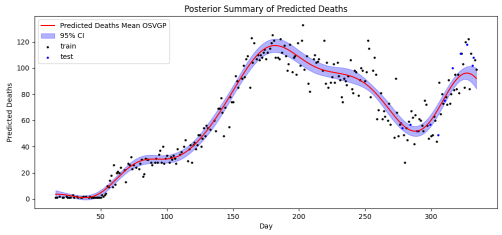


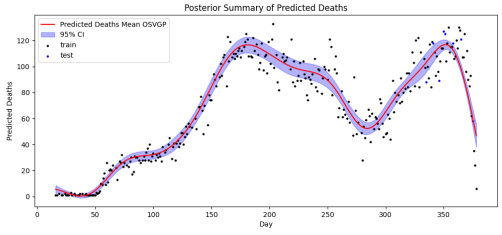
Figure 5. Task-pair comparison across the COVID-19 stream. The analytic construction closely tracks the ODE baseline in RMSE and R^2, improves NLPD on most evaluation pairs, and yields its clearest calibration gain around Task 4, while the final Eval 5 after Task 5 pair remains the main counterexample.



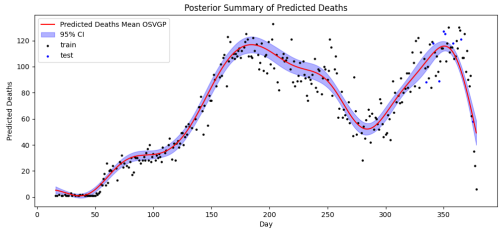
(a) Task 4, ODE



(b) Task 4, Analytic



(c) Task 5, ODE



(d) Task 5, Analytic

Figure 6. Representative COVID-19 posterior comparisons in a two-column layout. Task 4 illustrates a case where the analytic construction slightly improves predictive density while largely preserving the posterior shape of the ODE baseline, whereas Task 5 shows the main final-stage counterexample where the analytic variant underperforms.